SAT Elbek
SAT PREPARATION

DIGITAL SAT · FULL-LENGTH PRACTICE

October Full Practice Test

Reading and Writing · Math · Adaptive second modules with answer key

SECTION 1

Reading and Writing

Module 1 · Baseline · 32 questions
Module 2 · Easier · 29 questions
Module 2 · Harder · 30 questions

SECTION 2

Math

Module 1 · Baseline · 24 questions
Module 2 · Easier · 25 questions
Module 2 · Harder · 27 questions

How to take this test

1. Begin each section with Module 1 (Baseline).
2. Based on your Module 1 performance, continue with either the Easier or the Harder Module 2.
3. Check your work with the Answer Key at the end of this booklet.

STUDENT NAME _____ DATE _____ SCORE _____

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Section 1, Module 1: Reading and Writing

Baseline • 32 questions

Part 1: Vocabulary

Words in context

1 Question 1

VOCABULARY

_____ the source of a malfunction in a complex electrical system can be a painstaking, time-consuming process: a repair technician may need to test each individual component one by one until the specific part causing the problematic behavior is identified.

Which choice completes the text with the most logical and precise word or phrase?

A Adapting

B Isolating

C Incorporating

D Fabricating

2 Question 2

VOCABULARY

Single-celled foraminifera inhabiting coastal waters experience significant environmental salinity changes during tidal cycles. How, then, do they _____ internal salt levels? Some organisms maintain high concentrations of phosphates that allow them to affect the concentration of internal electrolytes—including salt—in response to external changes. It may be that coastal foraminifera do the same.

Which choice completes the text with the most logical and precise word or phrase?

A counteract

B elude

C tolerate

D regulate

3 Question 3

VOCABULARY

Though most studies of the effect of altitude on blood chemistry usually concentrate on people who live above sea level, researchers Suleiman A. Al-Sweedan and Moath Alhaj have instead chosen the _____ path in their recent work of studying the blood of people who live below sea level, in locations such as the California towns of Westmorland and Thermal.

Which choice completes the text with the most logical and precise word or phrase?

A predictable

B innovative

C idealistic

D timeworn

4 Question 4

VOCABULARY

While once _____, basic computer security training that only teaches recognition of obvious scams cannot provide enough protection against today's sophisticated fake emails, which employ such subtle deception techniques that many organizations now require their staff to complete comprehensive cybersecurity education programs.

Which choice completes the text with the most logical and precise word or phrase?

A uncommon

B superficial

C sufficient

D affordable

5 Question 5

VOCABULARY

The architectural firm's design for the new municipal library was deliberately _____; by rejecting ornate embellishments in favor of clean lines and unadorned surfaces, the architects prioritized utility and simplicity over grandeur.

Which choice completes the text with the most logical and precise word or phrase?

A austere

B capricious

C ostentatious

D convoluted

Part 2: Reading

Structure and purpose, cross-text connections, central ideas, evidence, and inferences

6 Question 6

READING

What is a city? The answer depends on where you live! Many countries define an area as a city based on how many people live there. However, not every country uses the same numbers. Iceland defines a city as an area with a population of at least 200, while the Netherlands defines a city as having a minimum population of 20,000. Some countries even define cities using other factors, like if there is a local government.

Which choice best describes the overall structure of the text?

A. It lists the number of cities in one country, then compares that number to the number of cities in another country.

B. It poses a question, then explains why the question has many answers.

C. It offers a recommendation, then provides reasons for that recommendation.

D. It describes two locations, then explains why only one of the locations is considered a city.

7 Question 7

READING

The following text is adapted from the 1940 novel *The Hamlet* by William Faulkner. In rural Mississippi, a man is riding a horse toward the residence of a man he intends to meet about a business matter.

> He was breathing hard but not because he had been galloping. On the contrary, since he had approached near enough to his destination to believe he could have seen smoke if there had been smoke, he had ridden slower and slower. Nevertheless he now sat the horse before the gap in the fence, breathing hard enough through his nose and even sweating a little, looking at the sagging broken-backed cabin set in its inevitable treeless and grassless plot and weathered to the color of an old bee-hive, with that expression of tense and rapid speculation of a man approaching a dud howitzer [artillery] shell.

Which choice best states the main purpose of the text?

A. To contrast the man’s desire to attend the meeting with his awareness of the risk involved in attending it

B. To compare the upcoming meeting to a far more serious experience from earlier in the man’s life

C. To depict the man’s insistence on meeting despite the inconvenience of traveling on horseback

D. To portray the man’s anxiety about the meeting as reflected by his physical state and demeanor

8 Question 8

READING

Modern forestry offers diverse career opportunities that integrate scientific knowledge with hands-on environmental work. Arborists assess tree health and perform maintenance such as pruning to promote safety and prevent problems. Forestry technicians collect data on growth patterns and biodiversity in the wild. Forest ecologists conduct field research and analyze ecosystems to understand how forests respond to changes in weather patterns. With this blend of scientific understanding and practical application, forestry professionals address a variety of complex environmental challenges.

Which choice best describes the main purpose of the text?

A. To explain technological advances in modern forestry and demonstrate how they've transformed traditional practices

B. To analyze environmental challenges facing forests and describe how modern professionals address these problems

C. To compare career opportunities in modern forestry and explain which positions offer the best job security

D. To highlight career opportunities in modern forestry and describe how they integrate scientific knowledge and hands-on work

9 Question 9

READING

James Fong and his team claim to have discovered a new color, which they named olo. Olo is a very saturated blue-green color that can’t be seen with the human eye under natural conditions. In the human eye, long, medium, and short cones in the retina help sense bright light and color. Olo can only be seen by activating the medium cones alone. Fong and team activated these cones by using a targeted laser technology to stimulate just the medium cone cells.

According to the text, how can the color olo be seen?

A. By activating only the medium cones in the retina

B. By activating the long, medium, and short cones in the retina

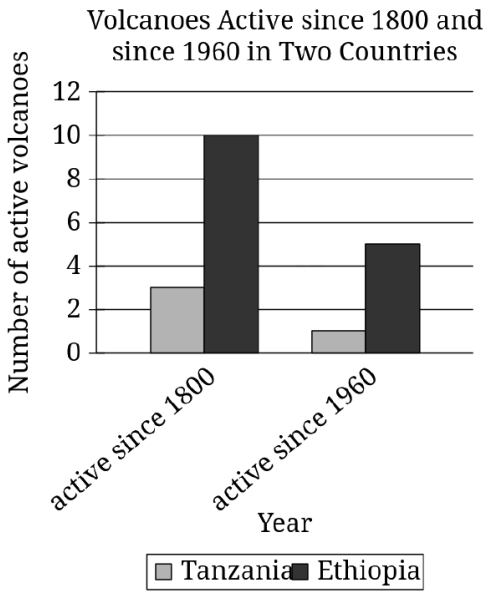
C. By activating both the long and short cones in the retina

D. By activating only the long cones in the retina

10 Question 10

READING

A student is collecting data about countries with volcanoes that have been active in the time since 1800 and in the time since 1960. The student records that Ethiopia has _____



Which choice most effectively uses data from the graph to complete the statement?

A) the same number of volcanoes that have been active since 1800 as Tanzania has.

B) no volcanoes that have been active since 1800.

C) more volcanoes that have been active since 1960 than Tanzania has.

D) fewer volcanoes that have been active since 1960 than Tanzania has.

11 Question 11

READING

Editor Jared Shurin's enormous 2023 anthology *The Big Book of Cyberpunk* comprises 108 stories. Shurin's volume is chronologically expansive, featuring such stories as James Tiptree Jr.'s "The Girl Who Was Plugged In" (1973) and Erica Satifka's "Act of Providence" (2021). In his introduction, Shurin defines cyberpunk as a subgenre of speculative fiction concerning "the influence of technology on the scale, the pace, or the pattern of human affairs"; however, some critics assert that cyberpunk is a literary movement that is in large part aesthetic rather than idea driven.

Which quotation from a literary critic would be the most effective evidence in support of the underlined claim?

A "Not all the stories anthologized in *The Big Book of Cyberpunk* were first published as stand-alone short stories in magazines or journals: for example, qntm's 2021 story 'Lena' was originally self-published on the author's site qntm.org."

B "The stories in *The Big Book of Cyberpunk* share a number of common themes indicative of cyberpunk as a whole: body modification; resistance to authority; the ways in which technological advancements alter human behavior."

C "Shurin's characterization of cyberpunk is remarkably generous: it is a categorization that might well serve for the entire field of science fiction in general, rather than cyberpunk in particular."

D "A key component of cyberpunk that differentiates it from other subgenres of science fiction is an affection for the totems of 1980s pop culture, such as the mirrored sunglasses that were then popular."

12 Question 12

READING

Tribal nations employ a large percentage of their populations in tribally owned businesses as well as in government and public administration. This is especially true in Oklahoma and other states with robust tribal economies and large Native populations. Between end-of-year 2017 and end-of-year 2023, the percentage of tribal employees working in the public sector increased, from 34.0% to 34.6%, while the percentage working in the business sector declined from 66.0% to 65.4%.

Employment in Tribal Economies in December 2017 and December 2023 by Sector

Economic sector	Total employees, 2017	% of workforce, 2017	Total employees, 2023	% of workforce, 2023
Business sector (all tribal businesses)	220,700	66.0%	232,300	65.4%
Leisure and hospitality businesses	185,100	55.3%	192,300	54.2%
Education and healthcare businesses	19,100	5.7%	21,600	6.1%
Other businesses	16,500	5.0%	18,400	5.1%
Public sector (tribal government and public administration)	113,800	34.0%	122,700	34.6%

Which statement about the decline in the percentage of tribal employees working in the business sector is best supported by information in the table?

A) It didn’t result from an overall contraction of that sector, since the total number of employees in that sector increased by several thousand during this period.

B) It was offset by changes in leisure and hospitality businesses, since the percentage of employees working in those businesses increased during this period.

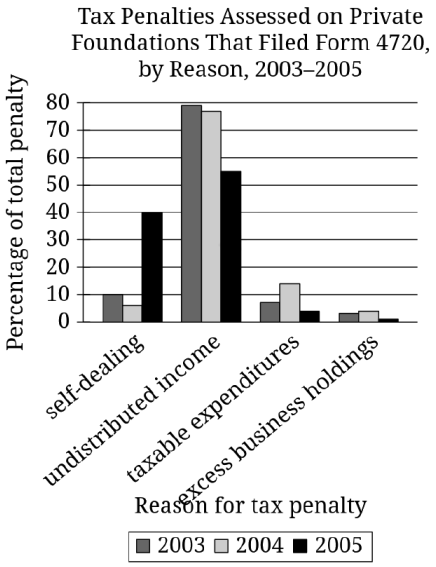
C) It was driven primarily by changes in businesses classified as “other,” since those businesses had the lowest number of employees by the end of 2023.

D) It resulted from widespread workforce redistributions, since several thousand employees left that sector to take new jobs in the public sector during this period.

13 Question 13

READING

While US public charities, like Commonfund, must file Form 990 yearly with the IRS, private foundations, such as Lilly Endowment Inc., must file a different form, 990-PF. In addition, foundations that engage in certain prohibited activities must also file Form 4720 and pay an excise tax on the money involved. Private foundations are prohibited from holding excess interests in a business enterprise, “self-dealing” (conducting activities that benefit foundation insiders), making taxable expenditures such as grants to noncharitable entities, and failing to cross a required threshold in making charitable distributions from income. Out of the organizations that filed Form 990-PF in the years 2003–2005, _____



Which choice most effectively uses data from the graph to complete the assertion?

A. those that filed Form 4720 because they had excess holdings in a business enterprise paid, on average, a larger penalty than those that filed Form 4720 because they made taxable expenditures.

B. those that filed Form 4720 because they engaged in self-dealing paid an increasing percentage of the total penalty each year from 2003 to 2005.

C. those that filed Form 4720 in 2003 because they failed to meet the minimum charitable distribution requirement collectively paid a larger percentage of the total penalty assessed that year than those that filed Form 4720 in 2005 for the same reason.

D. those that filed Form 4720 in 2005 because they engaged in self-dealing collectively paid a larger percentage of the total penalty assessed that year than those that filed Form 990 but also filed Form 4720 for the same reason.

14 Question 14

READING

Unlike terrestrial and shallow-water ecosystems that rely predominantly on photosynthetic primary producers, hydrothermal vent communities depend on chemolithoautotrophic bacteria. These microscopic organisms derive metabolic energy by oxidizing hydrogen sulfide emitted from geothermal fissures along the ocean floor, forming the bedrock of an abyssal food web isolated from solar radiation.

Based on the text, what distinguishes hydrothermal vent communities from surface ecosystems?

A. Their reliance on chemical oxidation rather than sunlight for primary energy generation.

B. Their complete vulnerability to seasonal shifts in solar radiation levels.

C. Their exclusive dependence on plant matter descending from shallow waters.

D. Their inability to maintain complex ecological interactions in cold temperatures.

15 Question 15

READING

Philoctetes is a play by Sophocles that was first performed in 409 BCE. In the play, Odysseus and Neoptolemus must retrieve Philoctetes—a master archer who is believed necessary for the Greeks to win the Trojan War—from a cave on a deserted island where Odysseus and others had abandoned him with a debilitating injury earlier in the war. Odysseus suggests that the plan for returning with Philoctetes rests on Neoptolemus not having been part of the group that abandoned him: _____

Which quotation from *Philoctetes* most effectively illustrates the claim?

A. "Since the prize of victory is so dear, / Endure it [acting deceitfully]—We'll be just another day / But now, for one brief hour, devote thyself / To serve me without shame."

B. "Go near now silently, and make me know / If still he persevere, and hold this spot, / Or have roamed elsewhere, that informed of this / I may proceed with what remains to say, / And we may act in concert."

C. "Now let me show, why thine approach to him / Is safe and trustful as mine cannot be / Thou didst sail forth, not to redeem thine oath, / Nor by constraint, nor with the foremost band. / All which reproaches I must bear."

D. "No moment for this talk, / Lest he discern my coming, and I lose / The scheme, wherewith I think to catch him soon."

16 Question 16

READING

Louis-Ferdinand Céline's 1932 novel *Journey to the End of the Night* is regularly described as autobiographical. That characterization is apt—there are many parallels between the experiences of the novel's protagonist, Ferdinand Bardamu, and those of Céline—but it should not be taken to mean that every person or event depicted in *Journey to the End of the Night* has a real-life analogue. Much of the novel is pure invention, and readers who ignore this fact and instead focus excessively on correspondences between the novel and Céline's life can thus

_____

Which choice most logically completes the text?

A) fail to appreciate how much creativity Céline shows in the book.

B) overemphasize the extent to which Céline took inspiration from earlier writers.

C) refuse to acknowledge that Céline drew on real-world material when writing *Journey to the End of the Night*.

D) misrepresent *Journey to the End of the Night* as being more widely read than it actually is.

17 Question 17

READING

Last-mile delivery, the final stage in the delivery of products to customers, is subject to increasing pressure as consumers place greater emphasis on convenience and immediacy, often expecting next-day delivery. In response to this challenge and others, such as the increasing volume of parcels to be delivered, delivery companies are continually looking for ways to make last-mile delivery more efficient. One advance in last-mile delivery is the use of self-service lockers (containers where consumers can pick up their parcels using a unique code). This approach, while promising, also introduces new obstacles (e.g., convenient placement of self-service lockers is challenging). It is therefore likely that _____

Which choice most logically completes the text?

A) delivery companies will need to keep experimenting with new ideas for last-mile delivery if they hope to keep pace with consumers' expectations.

B) delivery companies will continue promising next-day delivery out of a desire to meet consumers' expectations.

C) advances in last-mile technologies will slow as delivery companies focus more on meeting customer expectations for increased convenience.

D) innovations in the use of self-service lockers for last-mile delivery will stall unless delivery companies shift their focus to methods with fewer complications.

18 Question 18

READING

A cognitive psychology study revealed that participants memorizing pairs of abstract geometric symbols exhibited 35% higher retention when tested following eight hours of sleep compared to eight hours of wakefulness. If consolidation of non-verbal, abstract information occurs predominantly during deep slow-wave sleep cycles, it can be logically inferred that _____

Which choice best completes the text based on the reasoning presented?

A) wakefulness permanently disables the neurological circuits that process abstract geometric forms.

B) slow-wave sleep plays a critical functional role in solidifying the neurological retention of abstract visual data.

C) sleep deprivation enhances memory retention for linguistic information rather than visual symbols.

D) abstract memory storage is completely dependent on intentional waking mental rehearsal.

19 Question 19

READING

Ecologist Dr. Elena Vance observed that urban wetlands filter pollutants at rates comparable to modern treatment facilities. However, Vance cautions that overloading these fragile ecosystems with industrial runoff can degrade their microbial communities, ultimately compromising their filtration efficiency. Without strict municipal regulations on chemical discharge, the natural buffering capacity of such wetlands could permanently diminish.

Which choice best describes the primary function of the underlined sentence in the overall structure of the text?

A) It introduces a technical method for measuring chemical pollutants in urban watersheds.

B) It qualifies the earlier praise of wetlands by noting a condition under which their benefits are compromised.

C) It dismisses artificial wastewater treatment facilities as ineffective alternatives.

D) It refutes a prevailing hypothesis regarding microbial diversity in aquatic environments.

Part 3: Grammar

Standard English conventions, transitions, and rhetorical synthesis

20 Question 20

GRAMMAR

With an average summer temperature of 82.5 degrees Fahrenheit, Nicosia is one European _____ many travelers recommend, especially from June to September.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) city, which

B) city in which,

C) city that

D) city, that

21 Question 21

GRAMMAR

Nanoengineers have developed synthesized mesoporous silica nanoparticles that are used to transport genes and medicines to specific cells in the body. As nanotechnology evolves, medical technology _____ to evolve as well.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had likely continued

B) would likely have continued

C) had likely been continuing

D) will likely continue

22 Question 22

GRAMMAR

To more easily balance the weight of her body on her toes while performing the demanding pointe work that became increasingly common in early 20th-century ballet, internationally known ballerina Anna Pavlova redesigned her pointe shoes. The modifications she made—adding a stiffened leather sole and hardened toe _____ began the modernization of ballet's iconic footwear.

Which choice completes the text so that it conforms to the conventions of Standard English?

A box,

B box—

C box

D box:

23 Question 23

GRAMMAR

Archaeological evidence indicates that board games have existed in human societies for at least five millennia, with examples discovered in the tombs of ancient Egypt and the ruins of Mesopotamia. The purpose of these ancient games was _____ provide entertainment during leisure time, to reflect cultural beliefs, and to develop strategic thinking skills among players.

Which choice completes the text so that it conforms to the conventions of Standard English?

A threefold. To

B threefold: to

C threefold; to

D threefold to

24 Question 24

GRAMMAR

Extensive radiocarbon dating of reed samples from Caral confirmed Ruth Shady Solís's hypothesis about the Peruvian site's _____ nearly 5,000 years ago, the ancient city preceded the Olmec civilization to the north by a full millennium, a revelation that fundamentally altered our understanding of urban development in the Western Hemisphere.

Which choice completes the text so that it conforms to the conventions of Standard English?

A age, built

B age: built

C age and built

D age built

25 Question 25

GRAMMAR

Chemists investigating the properties of coal tar in the mid-nineteenth century discovered compounds that led to the creation of vibrant artificial colorants. Commercial applications of this new synthetic dye technology—beginning with the purple dye called mauve, discovered by William Perkin in 1856—_____ responsible for launching what one science historian calls a "manufacturing revolution based on chemistry."

Which choice completes the text so that it conforms to the conventions of Standard English?

A was

B is

C were

D has been

26 Question 26

GRAMMAR

How does a clear evening become foggy? First, the ground releases heat, which radiates upward. This causes the air near the ground to cool. _____ tiny water droplets begin to form as the cooling air reaches its dew point and can no longer hold its moisture as invisible vapor. Finally, as more droplets accumulate, a visible layer of fog develops close to the ground.

Which choice completes the text with the most logical transition?

A. Similarly,

B. Lastly,

C. Instead,

D. Next,

27 Question 27

GRAMMAR

Scientists studying forest restoration have discovered that trees depend on complex partnerships with microscopic organisms for survival. The researchers found that special fungi called endophytes live inside tree leaves and serve as a defense mechanism. _____ these endophytes help by fighting off diseases and boosting the tree's immune system.

Which choice completes the text with the most logical transition?

A. In conclusion,

B. Nevertheless,

C. Alternatively,

D. Specifically,

28 Question 28

GRAMMAR

While researching a topic, a student has taken the following notes:

- Pistol shrimp are small marine crustaceans that hunt using a specialized oversized claw.
- When the claw snaps shut rapidly, it pushes water through a narrow groove at extreme speed.
- This high-speed water jet creates a bubble that pops forcefully near the shrimp's prey.
- The popping bubble generates shock waves powerful enough to stun small fish.

The student wants to introduce pistol shrimp to an audience unfamiliar with these crustaceans. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Some crustaceans use a specialized claw to create high-speed water jets and a bubble that stuns the small fish they prey on.

B. Pistol shrimp are small marine creatures that use a specialized oversized claw to generate shock waves powerful enough to stun small fish.

C. Pushing water through a narrow groove at extreme speeds, the snapping shut of the shrimp's claw sets the shock-wave hunting process in motion.

D. Pistol shrimp are preyed on by small marine crustaceans that create high-speed water jets to stun their prey.

29 Question 29

GRAMMAR

While researching a topic, a student has taken the following notes:

- Elizabeth Catlett (1915–2012) was a celebrated African American artist.
- She is best known for creating sculptures and prints that explore the Black experience.
- *Standing Mother and Child* is a 1980 sculpture by Catlett.
- *New Generation* is a 1992 print by Catlett.

The student wants to provide an example of one of Catlett's sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. The print *New Generation* was created by celebrated artist Elizabeth Catlett in 1992.

B. The sculpture *Standing Mother and Child* was created by celebrated artist Elizabeth Catlett in 1980.

C. Artist Elizabeth Catlett is best known for creating sculptures and prints that explore the Black experience.

D. Elizabeth Catlett, a celebrated artist, was born in 1915.

30 Question 30

GRAMMAR

While researching a topic, a student has taken the following notes:

- Chapantongo is a municipality in the state of Hidalgo, Mexico.
- Municipalities are governmental regions responsible for providing many public services to their residents.
- One service they provide is traffic control.
- Chapantongo's population was 12,967 in 2020.
- Its population was 12,271 in 2010.
- Hidalgo is divided into 84 municipalities.

The student wants to emphasize the 2020 population of Chapantongo. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. As of 2020, Chapantongo was one of 84 municipalities in Hidalgo, Mexico.

B. In 2020, the municipality of Chapantongo had a population of 12,967.

C. As a municipality in Hidalgo, Chapantongo is responsible for providing public services to its population.

D. Home to 12,271 people, the governmental region was responsible for providing traffic control to its population in 2010.

31 Question 31

GRAMMAR

While researching a topic, a student has taken the following notes:

- *The Future of Nostalgia* is a scholarly book by literary theorist Svetlana Boym.
- The book explores the concept of nostalgia from various angles.
- Chapter 8 explores nostalgia and Europe's largest shopping mall.
- Chapter 17 discusses various skeptics' takes on the concept of nostalgia.
- In chapter 17, Boym writes, "The poethics of nostalgia combines estrangement and human solidarity, affect and reflection."

The student wants to explain what Svetlana Boym writes about in chapter 8. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Svetlana Boym's *The Future of Nostalgia* explores the concept of nostalgia from various angles.

B. As Svetlana Boym writes in the seventeenth chapter of her book, "The poethics of nostalgia combines estrangement and human solidarity, affect and reflection."

C. Literary theorist Svetlana Boym explores nostalgia and Europe's largest shopping mall in chapter 8 of her book.

D. After exploring nostalgia and Europe's largest shopping mall, Svetlana Boym goes on to discuss various skeptics' takes on the concept of nostalgia.

32 Question 32

GRAMMAR

For decades, historical linguists hypothesized that Proto-Indo-European originated in the Pontic-Caspian steppe roughly 6,000 years ago. _____, recent ancient genomic analyses combined with phylogenetic linguistic modeling have provided compelling evidence supporting an earlier agricultural origin south of the Caucasus.

Which choice completes the text with the most logical transition?

A In addition

B However

C Consequently

D For example

Section 1, Module 2: Reading and Writing

Easier • 29 questions

Part 1: Vocabulary

Words in context

1 **Question 1** VOCABULARY

The following text is from Jacqueline Woodson’s 2018 novel *Harbor Me*.

> I’ve pulled the voice recorder from my closet and have it sitting on the middle of my bed now. When I press play, Esteban’s voice fills my room.

As used in the text, what does the word “pulled” most nearly mean?

A. Developed

B. Attracted

C. Startled

D. Taken

2 Question 2

VOCABULARY

The National Heritage Fellowship was created to honor exceptional folk and traditional artists in the United States. One artist who received the fellowship is Mexican American rodeo tailor Manuel Cuevas. Cuevas was chosen for his lifetime _____ the arts.

Which choice completes the text with the most logical and precise word or phrase?

A imitations of

B misunderstandings of

C contributions to

D doubts about

3 Question 3

VOCABULARY

Car manufacturers increasingly label certain vehicle components as “lifetime” parts that supposedly never need replacement. Mechanics tend to greet such assertions _____, as ample experience has shown them that environmental factors, varied driving conditions, and intensive use will likely lead to the eventual failure of these components.

Which choice completes the text with the most logical and precise word or phrase?

A unreasonably

B hastily

C skeptically

D ambivalently

4 Question 4

VOCABULARY

From fountains like *Crown Fountain* by Jaume Plensa, found at Millennium Park, to street art like Don’t Fret’s mural *Gift To The Class of 2016* on South Wabash Avenue, Chicago offers an array of works to _____ the tastes of art lovers.

Which choice completes the text with the most logical and precise word or phrase?

A mitigate

B venerate

C satiate

D supplant

Part 2: Reading

Structure and purpose, cross-text connections, central ideas, evidence, and inferences

5 Question 5

READING

Ethicist Ingrid Robeyns argues in her work that taxes on the very wealthy should be increased so that governments will be enabled to solve problems of collective action, such as mitigating climate change. Robeyns cites the work of economist Mariana Mazzucato, who shows that government subsidies are a more effective engine of innovation than private sector investments; in addition, government-funded innovations tend to lead to further developments in disparate fields (such as the United States mission to land a person on the moon, which accelerated developments in nutrition as well as communications).

Which choice best describes the overall structure of the text?

A. It summarizes the research of one scholar, then presents a study by another scholar that was inspired by the findings of that research.

B. It introduces a proposition by a scholar, then describes that scholar's use of findings by another scholar to support that proposition.

C. It details the findings of a research project by one scholar, then puts forward the suggestions of another scholar for further investigation.

D. It offers a claim by a scholar, then uses the results of a project by another scholar to show why that claim should be rejected.

6 Question 6

READING

Text 1

The novels of Henry James changed stylistically starting in 1902, becoming more demanding and featuring a much more measured pace than his earlier works. This places a burden on readers that they may not be prepared for: James's late novels are some of the best he wrote, but they are not for everyone.

Text 2

The films of Martin Scorsese released starting in 2016 have the same place in his filmography that the late novels of Henry James have in his bibliography. Both are works by proven masters—James had published *The American* (1877); Scorsese had directed *Mean Streets* (1973)—who began to make demanding works, such as James's *The Wings of the Dove* (1902) and Scorsese's *Silence* (2016), that ask more of their audiences than their earlier works do but give more in return.

Based on the texts, how would the author of Text 2 most likely respond to the underlined claim in Text 1?

A. By arguing that James's novels cannot be fully understood without also considering Scorsese's films

B. By conceding that James's *The American* is a superior novel to *The Wings of the Dove*

C. By asserting that both James and Scorsese ought to be considered masters of their chosen art forms

D. By agreeing that James's late novels are excellent but can be challenging for their readers

7 Question 7

READING

The following text is from Hilda Eunice Burgos's 2018 novel *Ana María Reyes Does Not Live in a Castle*. The narrator is in sixth grade.

> The next day I went to my piano lesson, as I did every Tuesday after school. My piano teacher, Doña Dulce Sánchez, lived two blocks from my school, in the opposite direction from my house.

According to the text, where did the narrator's piano teacher live?

A. Near the center of town

B. Two blocks from the narrator's school

C. Across the street from the narrator's home

D. One block from the narrator's home

8 Question 8

READING

Emulsification is a process that creates a uniform mixture that includes two ingredients that do not combine on their own, such as oil and water. For this process to happen, a third ingredient, an emulsifier, must be combined with the other two. Carrageenan, a substance extracted from red seaweed, is a common emulsifier: it helps to create a smooth texture in condensed milk that makes it more pleasant to drink.

According to the text, what result is associated with carrageenan?

A. It helps ensure that condensed milk has a smooth texture.

B. It prolongs the time it takes for ingredients to mix together during the making of condensed milk.

C. It decreases the amount of one ingredient in condensed milk while increasing the amount of another.

D. It assists in the evaporation of water during the making of condensed milk.

9 Question 9

READING

Biologist Nathalie Stroeymeyt and her team found that black garden ants demonstrated a sophisticated form of “architectural immunity.” When confronted with fungal pathogens, these ants modified their excavation patterns in several strategic ways to speed up nest growth, increase spacing between entrances, and reduce the centrality of key chambers (containing the queen, food, or nursery). The team noted that these modifications weren’t random but specifically arranged to minimize fungal spread.

Which choice best states the main idea of the text?

A. A team of scientists noticed that fungal pathogen exposure causes black garden ants to abandon their existing nest structures.

B. According to a team of biologists, black garden ants build larger nests when exposed to a harmful fungus than they otherwise would build.

C. A study found that black garden ants deliberately modify their nest architecture to keep a harmful fungus from spreading.

D. A study noted that black garden ants exposed to a harmful fungus tend to dig longer tunnels in their nests instead of reducing the centrality of the nest’s chambers.

10 Question 10

READING

Peter Pan is a 1911 novel by J.M. Barrie. In the fantasy novel, a young boy named Peter lives on a mythical island. The narrator indicates that Peter does not know that pretend things are not real: _____

Which quotation about Peter from *Peter Pan* best supports the claim?

A. “He boasted that he had gone walking for the good of his health.”

B. “He had fallen at once into a dreamless sleep.”

C. “In his absence things are usually quiet on the island.”

D. “To him make-believe and true were exactly the same thing.”

11 Question 11

READING

A student is researching the Amstrad CPC and other important gaming systems that were part of the global rise of the home video game industry during the 1970s and 1980s. The student is surprised to find that the Amstrad CPC sold relatively few units worldwide, with only about _____

Home Video Game Systems of the 1970s and 1980s

System	Manufacturer	Approximate number of units sold worldwide
Apple II	Apple Inc.	4,487,000
Amstrad CPC	Amstrad	2,000,000
Game & Watch	Nintendo	18,600,000
Intellivision	Mattel	3,000,000

Which choice most effectively uses data from the table to complete the statement?

A) 2,000,000 units sold compared to the approximately 18,600,000 units sold of the Game & Watch.

B) 3,000,000 units sold compared to the approximately 4,487,000 units sold of the Game & Watch.

C) 2,000,000 units sold compared to the approximately 18,600,000 units sold of the Intellivision.

D) 3,000,000 units sold compared to the approximately 4,487,000 units sold of the Apple II.

12 Question 12

READING

Brittle stars are sea animals that move along the ocean floor using their five arms. Unlike most animals, brittle stars have 70%–80% of their weight in their arms. To better understand the benefits of this unusual trait, scientists built a soft robot inspired by the brittle star's body structure. The robot has a middle disk and five arms whose weights can be changed. When the scientists made the arms heavier, the arms pressed down more firmly on the ground. This stronger grip kept the arms from slipping and allowed the robot to move more efficiently across the ocean floor. Based on these results, the scientists hypothesize that _____

Which choice most logically completes the text?

A. brittle stars have most of the weight in their arms because that is where they store nutrients.

B. brittle stars have most of their weight in their arms so they can scare off predators.

C. having most of their weight in their arms allows brittle stars to climb steep underwater cliffs.

D. having most of their weight in their arms helps brittle stars move across the ocean floor.

13 Question 13

READING

The state of Maine has classified the smallmouth bass as an invasive species that could harm some of the state's native species. But researchers Alejandro Camacho and Jason McLachlan have pointed out that Earth's climate is changing in ways that challenge such classifications. Climate changes may force animals from their current ranges. Climate changes may also create good habitats in areas where a species couldn't live previously. These observations suggest that _____

Which choice most logically completes the text?

A. Maine was previously home to some smallmouth bass but they were outcompeted by invading species.

B. it's useful at present for Maine to distinguish between invasive and native species in some instances but not in the case of the smallmouth bass.

C. labels like the one that Maine has applied to the smallmouth bass reflect environmental conditions that may not persist.

D. even if Earth's climate doesn't change in the way scientists predict, the smallmouth bass will likely establish itself in Maine.

Part 3: Grammar

Standard English conventions, transitions, and rhetorical synthesis

14 Question 14

GRAMMAR

A multifaceted constellation of environmental factors, including oceanic surface temperatures, tropospheric wind shear, and ambient atmospheric humidity, _____ the developmental trajectory and destructive potential of hurricanes before landfall.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) dictate

B) dictates

C) are dictating

D) have dictated

15 Question 15

GRAMMAR

In the tapestries *Blessing of Rain* and *Intended Vermillion*, Diné textile artist D.Y. Begay uses _____ and yellows extracted from cochineal insects from Peru as well as local hollyhocks, coreopsis, and chamisa plants gathered from her Tsélání homeland in Arizona.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) reds, greens,

B) reds greens,

C) reds greens

D) reds; greens,

16 Question 16

GRAMMAR

Although Spanish artist Pablo Picasso (1881–1973) is famous mainly for his paintings, he worked in other art forms as well. Picasso _____ experimenting with sculpture as a young man and eventually created hundreds of innovative pieces, many of which he kept in his home.

Which choice completes the text so that it conforms to the conventions of Standard English?

A having begun

B began

C beginning

D to begin

17 Question 17

GRAMMAR

Colonial American newspapers played an important role in protesting British taxes, such as those resulting from the Stamp Act in 1765. Publisher William Rind showed his opposition by printing skull and crossbones symbols _____ the front page of his newspaper, the *Maryland Gazette*, where tax stamps would normally go.

Which choice completes the text so that it conforms to the conventions of Standard English?

A on:

B on,

C on;

D on

18 Question 18

GRAMMAR

Rock carvings found on islands around the lower Susquehanna River in Maryland repeatedly feature the same stylized face. Experts are divided on whether the face represents the head of a _____ a human, or a mythical figure.

Which choice completes the text so that it conforms to the conventions of Standard English?

A fish: a serpent

B fish. A serpent,

C fish a serpent,

D fish, a serpent,

19 Question 19

GRAMMAR

In ancient China, the art of miniaturizing trees in containers was called “penjing.” This term, which means “potted scenery,” _____ in China.

Which choice completes the text so that it conforms to the conventions of Standard English?

A to still use

B still being used

C is still used

D still used

20 Question 20

GRAMMAR

A recent study _____ by biologists Brandi Pessman and Eileen Hebets at the University of Nebraska–Lincoln found that in noisy environments, some funnel-weaving spiders built webs that amplified the sound of incoming vibrations at a particular frequency, allowing the spiders “to better hear certain signals above the noise,” according to Pessman.

Which choice completes the text so that it conforms to the conventions of Standard English?

A. had been conducted

B. is conducted

C. conducted

D. was conducted

21 Question 21

GRAMMAR

Many cities invest in public art, displaying works in locations where they can be enjoyed by anyone. On display in Charlotte, North Carolina, _____ is a sculpture with rotating steel layers that align to form a human head.

Which choice completes the text with the most logical transition?

A. however,

B. nevertheless,

C. for instance,

D. finally,

22 Question 22

GRAMMAR

In a 1993 study by Vales and Peek, the researchers used microhistological fecal analysis to determine the ratio of three plant subtypes (graminoids, forbs, and browse) within the diets of North American ungulates. The researchers did not perform this analysis on all such ungulates, _____ they focused exclusively on elk in Wyoming.

Which choice completes the text with the most logical transition?

A) in other words;

B) regardless;

C) for instance;

D) however;

23 Question 23

GRAMMAR

Lalo Schifrin, an Argentine American composer and pianist who wrote and recorded a number of jazz albums with trumpeter Dizzy Gillespie in the 1960s, also composed music for many films and television shows. _____ Schifrin's most iconic work, the theme for the 1966 TV show *Mission: Impossible*, uses a jazz-inspired 5/4 time signature.

Which choice completes the text with the most logical transition?

A) Conversely,

B) In other words,

C) Fittingly,

D) Actually,

24 Question 24

GRAMMAR

In Angola, the National Assembly is elected via a proportional representation (PR) system. In PR elections, votes are cast (not for specific candidates, as they are in single-member plurality systems, but for political parties) and then tabulated; _____ each qualifying party is awarded a number of seats proportional to the number of votes it received.

Which choice completes the text with the most logical transition?

A) accordingly,

B) by contrast,

C) second of all,

D) in fact,

25 Question 25

GRAMMAR

One might assume that the Netherlands' Gemini offshore wind farm produces the same amount of energy as land-based wind farms of a similar size. _____ because wind speeds over open water are generally higher than those over land, Gemini's turbines turn faster and produce more energy than do most land farms' turbines.

Which choice completes the text with the most logical transition?

A) Finally,

B) Therefore,

C) For example,

D) Actually,

26 Question 26

GRAMMAR

- The Highpointers Club is a hiking club.
- One of the main goals among club members is to reach the highest points in all fifty US states.
- Those who achieve this are called 50 Completers.
- In Suk Han became a 50 Completer on November 15, 2013.
- The highest point in Wisconsin is Timms Hill, at 1,951 ft.
- The highest point in Wyoming is Gannett Peak, at 13,809 ft.

Which choice most effectively uses information from the given sentences to explain the 50 Completers hiking challenge to a new audience?

A. If you are looking for a new hiking challenge, consider joining the Highpointers Club, as did In Suk Han, a hiker who successfully reached the highest point in every US state.

B. Not until after you have reached the highest points in all fifty US states—including Timms Hill in Wisconsin and Gannett Peak in Wyoming—can you include yourself among the 50 Completers of the Highpointers Club.

C. On November 15, 2013, In Suk Han finally completed the feat of reaching the highest point in all fifty US states, including Timms Hill in Wisconsin and Gannett Peak in Wyoming.

D. Hikers aiming to count themselves among the 50 Completers must reach not only Wisconsin's Timms Hill but also the even higher peak of Gannett Peak in Wyoming.

27 Question 27

GRAMMAR

- Cinematographers work with cameras and lighting.
- They help translate film directors' ideas into visual images.
- *Passengers* (2016) was directed by Morten Tyldum.
- Rodrigo Prieto was the film's cinematographer.

Which choice most effectively uses information from the given sentences to provide an example of a film Prieto worked on?

A. One example of Rodrigo Prieto's work as a cinematographer is the 2016 film *Passengers*.

B. As cinematographer, Rodrigo Prieto works with cameras and lighting to translate the film director's ideas into visual images.

C. Cinematographer Rodrigo Prieto and director Morten Tyldum have worked together.

D. In filmmaking, directors work with others to translate their ideas into the visual images that we encounter on the screen.

28 Question 28

GRAMMAR

While researching asteroid 16 Psyche, a student gathered the following notes:

- Asteroid 16 Psyche orbits the Sun within the main asteroid belt between Mars and Jupiter.
- The majority of known asteroids are composed primarily of rocky silicate minerals, ice, or carbonaceous matter.
- Psyche is exceptionally dense and consists almost entirely of metallic iron and nickel.
- Scientists hypothesize that Psyche may be the exposed metallic core of an early protoplanet.

The student wants to emphasize the unique composition of asteroid 16 Psyche compared to other asteroids. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Orbiting between Mars and Jupiter, asteroid 16 Psyche was the target of an exploratory mission launched by NASA.

B. Unlike typical asteroids composed mainly of silicates, ice, or carbon, 16 Psyche consists almost entirely of metallic iron and nickel.

C. Most asteroids in the main belt contain rocky silicates, though scientists study multiple compositions to investigate planetary formation.

D. Scientists hypothesize that Psyche may represent the exposed core of an early protoplanet due to its high density.

29 Question 29

GRAMMAR

While researching a topic, a student has taken the following notes:

- *Waiting* is a 1987 black-and-white linocut print by Kuwaiti artist Thuraya Al-Baqsami.
- It depicts a tranquil, everyday scene (a woman in a headscarf looking out a window).
- *¡Sera toda nuestra!* (“It will all be ours!”) is a 1977 color linocut print by Mexican American artist Carlos Cortéz.
- It features a scene with an explicitly political point of view (a group of laborers preparing to go on strike).
- Relief printing is a technique in which an image is carved onto a printing block, covered in ink or paint, and stamped onto paper.
- Linocut is a type of relief printing that uses linoleum tile as the printing block.

The student wants to contrast the subject matter of the two prints. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. Cortéz made *¡Sera toda nuestra!* (“It will all be ours!”) in 1977, while Al-Baqsami made *Waiting* later, in 1987.

B. The scenes depicted in both works were first carved onto a printing block, then stamped onto paper; however, one work is a linocut, while the other is a relief print.

C. Cortéz’s print expresses an explicitly political point of view, while Al-Baqsami’s depicts a tranquil, everyday scene.

D. The print by Cortéz is a color linocut; by contrast, Al-Baqsami’s is a relief print.

Section 1, Module 2: Reading and Writing

Harder • 30 questions

Part 1: Vocabulary

Words in context

1 **Question 1** VOCABULARY

Weather patterns, market trends, input costs, labor availability, equipment maintenance, and biological cycles are all variables that farmers must consider separately and in terms of how they may affect each other, and it is this _____ of factors in particular that makes successful farm management so challenging.

Which choice completes the text with the most logical and precise word or phrase?

A) neglect

B) misinterpretation

C) resolution

D) interplay

2 Question 2

VOCABULARY

To maximize _____, package distribution centers employ advanced scheduling systems and strategic staging areas that allow for goods received from warehouses to be immediately transferred to trucks for delivery to their final destinations. Some centers are so efficient that even the largest delivery trucks can be filled in a matter of minutes.

Which choice completes the text with the most logical and precise word or phrase?

A throughput

B staffing

C overhead

D liability

3 Question 3

VOCABULARY

Researchers have developed algorithms that take into account how various _____ influence production schedules, allowing rapid solution generation for manufacturing planning problems. The algorithms help identify efficient production schedules by processing those quantifiable inputs (e.g., demand patterns and capacity constraints) without exhaustively testing all possible scenarios.

Which choice completes the text with the most logical and precise word or phrase?

A optimizations

B predictions

C parameters

D anomalies

4 Question 4

VOCABULARY

Marvel Girl’s ability to lift modest loads with her mind is illustrative of the lamentably lackluster powers bestowed on many female superheroes, but even those with more formidable powers have been subject to unflattering portrayals that often serve as _____: as the Phoenix, Jean Grey loses control of a cosmic force and destroys an entire planet, showing the corrupting influence of power.

Which choice completes the text with the most logical and precise word or phrase?

A venerations

B admonitions

C obfuscations

D exonerations

Part 2: Reading

Structure and purpose, cross-text connections, central ideas, evidence, and inferences

5 Question 5

READING

Research on minor solar eruptions in magnetically quiet regions revealed an interesting pattern. Material expelled during these events follows a narrow arc trajectory back to the Sun's surface. High-resolution images show these eruptions produce thin streams of hot gas moving at speeds between 55 and 200 kilometers per second. Although these are impressive speeds, they are not sufficient to overcome the confining forces of the Sun's gravity and magnetic field, so the material is pulled back down.

Which choice best describes the function of the underlined sentence in the text as a whole?

A. It describes a natural phenomenon that is explained later in the text.

B. It presents a research finding that challenges the view mentioned in the first sentence of the text.

C. It acknowledges an exception to the pattern described in the first sentence of the text.

D. It discusses a natural occurrence that leads to the consequences explained later in the text.

6 Question 6

READING

With examples ranging from Dean’s Blue Hole in the Bahamas to Watling’s Blue Hole near San Salvador Island, blue holes are large underwater sinkholes: pits filled with dark blue water that contrasts with lighter blue shallower waters surrounding them. The formation of blue holes typically involves several natural processes, including the dissolution of limestone or other soluble rocks. Currently, the deepest blue hole on record is the Taam Ja’ Blue Hole in Chetumal Bay, Mexico, which is reported by Laura Carrillo and colleagues as extending at least 420 meters below sea level.

Which choice best describes the overall structure of the text?

A. It describes the appearance of a particular type of sinkhole, compares the formations of two sinkholes of that type, and then names the researchers who discovered the deepest known sinkhole of that type.

B. It compares a deep sinkhole to a shallower one, details the processes by which both sinkholes formed, and emphasizes the remarkable depth of the deeper sinkhole.

C. It explains how a particular sinkhole formed, evaluates that sinkhole’s unusual appearance, and names a research team that is studying the sinkhole.

D. It defines a kind of sinkhole, describes its typical appearance and formation, and provides information about the deepest known sinkhole.

7 Question 7

READING

Text 1 is from George Eliot's 1860 novel *The Mill on the Floss*. Text 2 is from Rabindranath Tagore's 1921 nonfiction book *Glimpses of Bengal*.

Text 1

The wood I walk in on this mild May day, with the young yellow-brown foliage of the oaks between me and the blue sky, the white star-flowers and the blue-eyed speedwell and the ground ivy at my feet, what grove of tropic palms, what strange ferns or splendid broad-petalled blossoms, could ever thrill such deep and delicate fibres within me as this home scene?

Text 2

My feelings seem to be those of our ancient earth in the daily ecstasy of its sun-kissed life; my own consciousness seems to stream through each blade of grass, each sucking root, to rise with the sap through the trees, to break out with joyous thrills in the waving fields of corn, in the rustling palm leaves.

Which choice best describes a notable difference in how the references to palm trees function in Text 1 and Text 2?

A. While the narrator of Text 1 and the author of Text 2 both refer to metaphorical palm trees, only the narrator of Text 1 suggests that the palm trees are representative of home.

B. While palm trees form part of a series conveying the variety of the natural world in both texts, only the author of Text 2 recognizes how closely interconnected palm trees are with other plants in their environment.

C. While the narrator of Text 1 and the author of Text 2 both discuss palm trees to suggest the value of experiencing the natural world, only the narrator of Text 1 alludes to the labor that goes into cultivating real palm trees.

D. While palm trees serve to evoke natural beauty in both texts, only the author of Text 2 expresses a close sense of identification with palm trees.

8 Question 8

READING

Primarily encountered by modern readers as inert written texts, the vernacular riddles of early medieval England contain “implicit vestiges” of an interactive performance context, scholar Jill Frederick argues. Locative phrases (e.g., “here” and “now”) scattered throughout the riddles indicate the space shared by their speakers and audiences; the common refrain “say what I am called” invites a verbalized response; and the conflation of many of the riddles’ speakers with their solutions (typically animals, objects, or natural phenomena) suggests a form of dramatic impersonation perhaps meant to entertain during public gatherings.

Which choice best states the main idea of the text?

A. It is reasonable to conclude based on textual evidence from the vernacular riddles of early medieval England that they were mainly performed at large public gatherings, though documentation of the locations in which they were performed is unlikely to exist.

B. Linguistic evidence found within the vernacular riddles of early medieval England suggests that they may have been intended to be performed in communal settings, though this isn’t how they are generally experienced by readers today.

C. People of early medieval England likely found that era’s vernacular riddles to be more amusing than modern readers do, given that there is substantial evidence to indicate that they are best experienced in the context of an interactive performance rather than exclusively as written texts.

D. Scholars once assumed that the vernacular riddles of early medieval England were originally intended to be read silently, but newly uncovered historical documents corroborate Jill Frederick’s assertion that they may in fact have been intended primarily for oral performance.

9 Question 9

READING

In the Mexican dialect of Spanish, the consonant *s* is pronounced with greater tension in the tongue than it is in Castilian, the dominant dialect in Spain, resulting in a sharper sound. Margarita Hidalgo, a linguist at San Diego State University, argues that Mexican Spanish acquired this pronunciation from Nahuatl, the language of the Nahua people in central Mexico: as the Spanish colonizers of Mexico incorporated Nahuatl words into their speech, and as more and more Nahuas spoke Spanish as a second language, the Nahuatl pronunciation of the *s* consonant replaced the Castilian pronunciation.

Information in the text best supports which statement about the consonant *s*?

A. The pronunciation of this consonant is the principal difference between the Mexican and Castilian dialects of Spanish.

B. This consonant appears in a higher percentage of Nahuatl vocabulary than of Castilian vocabulary.

C. The pronunciation of this consonant in Nahuatl may have been affected by its pronunciation in Mexican Spanish.

D. The pronunciation of this consonant requires greater tension in the tongue in Nahuatl than in Castilian.

10 Question 10

READING

Cover crops, such as teff, are planted by many farmers between main growing seasons. While these crops are known for increasing biodiversity, Axel Don and his coauthor argued in a 2015 study that cover crops' role in supporting soil carbon sequestration deserves more attention. Cover crops remove carbon dioxide (CO_2) from the air and convert it to organic compounds; some of these compounds are stored in the soil as soil organic carbon (SOC), which contributes to the buildup of SOC stocks over time. Based on data from numerous farm fields, Don and his coauthor concluded that cover crops contribute meaningfully to soil carbon sequestration and, if widely adopted, could help offset agricultural emissions of atmospheric CO_2 and other greenhouse gases.

Which finding, if true, would be most consistent with Don and his coauthor's conclusion?

A. Teff captures considerably more CO_2 in fields where it is the sole cover crop compared with fields where it is grown alongside other species of cover crops.

B. Cover crops generally exhibit reduced carbon sequestration rates in fields exposed to high levels of greenhouse gas emissions compared with fields exposed to low levels of those emissions.

C. SOC stocks vary more in fields where cover crops have been grown for decades compared with fields where cover crops have only recently been introduced.

D. SOC stocks tend to be significantly lower in fields without cover crops compared with fields where cover crops have been grown for several years.

11 Question 11

READING

While the brain regions most strongly associated with language tasks have some modularity in that different language tasks activate these areas to varying degrees, comprehension also invokes other areas of the brain more strongly associated with nonlinguistic functions. For example, understanding a description of a person performing physical actions might call on parts of the listener's brain associated with sensory-motor functions, whereas a highly emotive account will likely activate areas associated with affective processing. Thus, beyond the areas most strongly associated with linguistic functions, _____

Which choice most logically completes the text?

A) the specific combination of brain regions involved in comprehending speech is likely contingent on the total context in which the speech is embedded.

B) the brain regions associated with sensory-motor functions are often involved in general comprehension, even in contexts that don't involve descriptions of physical movement.

C) the brain regions most strongly associated with affective processing activate whenever a listener believes an utterance has emotional connotations, even if the speaker did not intend those connotations.

D) the areas of the brain that activate when processing speech will typically include areas primarily responsible for nonlinguistic functions, although the exact combination of areas may be impossible to determine.

12 Question 12

READING

The Clarion-Clipperton Zone (CCZ) is an expanse of abyssal plain and seamounts (underwater mountains) between Hawai‘i and Mexico in which mining is permitted, but the area’s biodiversity is poorly understood. The vast majority of invertebrate species found in a recent survey of the CCZ were hitherto unknown to scientists, and sampling for animal life has been highly concentrated in the eastern part of the zone. Some species like *Psychropotes dyscrita* were identified both morphologically and molecularly (using DNA analysis), whereas *Discantenna metallica* and other species for which molecular sampling was impracticable were identified solely based on physical morphology. Since molecular analysis does not require distinguishing subtle physiological differences, it is _____

Which choice most logically completes the text?

A. more frequently used for identifying species with physiological characteristics like those of *Psychropotes dyscrita* than morphological analysis is.

B. capable of providing greater physiological detail about a species than is the method used for *Discantenna metallica*.

C. better suited to distinguishing *Discantenna metallica* from morphologically similar species than to distinguishing *Psychropotes dyscrita* from morphologically similar species.

D. more likely to properly distinguish between morphologically similar but distinct invertebrate species than is the method used to identify *Discantenna metallica*.

13 Question 13

READING

Text 1

Paleontologists historically hypothesized that sauropod dinosaurs held their long necks vertically aloft like modern giraffes, allowing them to browse undisturbed foliage at the tops of towering forest canopies.

Text 2

Biomechanical simulations conducted by Dr. Raymond Kent suggest that sauropod cervical vertebrae lacked the dorsal articulation required for sustained vertical posture. Kent asserts that sauropods instead swept their necks horizontally to forage broad swaths of low vegetation with minimal locomotion.

Based on the texts, how would the author of Text 2 most likely respond to the hypothesis discussed in Text 1?

A) By arguing that giraffes and sauropods share identical vertebral morphology.

B) By disputing the claim that sauropods sustained a predominantly vertical neck orientation.

C) By demonstrating that sauropods relied entirely on high-canopy gymnosperms for nutrition.

D) By showing that sauropod neck fossils are too degraded for accurate biomechanical modeling.

14 Question 14

READING

Rare species advantage (RSA) describes the situation in which there are few conspecifics (other members of the same species) in an organism's immediate surroundings, enabling the organism to avoid competition with conspecifics for the same resources; RSA is widely believed to play a crucial role in maintaining ecological stability. Among tree species in the Barro Colorado Island plot (latitude 9.15° N) and other tropical forest sites, there is little correlation between conspecific aggregation and the abundance of a given species in the wider local area, so RSA is expected, but in the Fenglin Nature Reserve (latitude 48.08° N) and other temperate forest sites, the aggregation-abundance correlation for tree species is strongly negative. Mathematical modeling suggests, however, that ecological niche distinction may mitigate the apparent risk to temperate ecological stability engendered by this pattern: _____

Which choice most logically completes the text?

A. as a tree species in temperate forests becomes more abundant, individuals of that species are more likely to be found near conspecifics, enabling abundant species to exploit localized resources to a greater extent than rare species can.

B. even though rare tree species in temperate forests tend to cluster with conspecifics more than rare tree species in tropical forests do, they are sufficiently rare locally that they benefit from RSA more than rare tropical species do.

C. although rare tree species in temperate forests encounter relatively high local competition from conspecifics, they can thrive by exploiting resources that are underexploited by other local species.

D. while an increase in abundance in one tree species often leads to a reduction in abundance of nearby species, temperate tree species tend to play different roles in the same ecosystem, reducing the expected effect of abundance on aggregation.

Part 3: Grammar

Standard English conventions, transitions, and rhetorical synthesis

15 Question 15

GRAMMAR

A premortem is an exercise in which teams imagine their project has already failed, then brainstorm possible reasons for that failure. Despite its apparent pessimism, _____ prompting teams to proactively develop mitigation strategies during the planning stage. As a result, the project is more likely to succeed.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) effective uncovering of potential problems by the team in this activity is done early,

B) this activity effectively uncovers potential problems early,

C) potential problems are effectively uncovered by this activity early,

D) teams effectively uncover potential problems early with this activity,

16 Question 16

GRAMMAR

The trailblazing achievements of Roger Bannister, the English track star famous for breaking the four-minute-mile barrier at Oxford's Iffley Road Track in 1954, _____ confined to the realm of athletics: Bannister also became a pioneering neurologist in the field of autonomic disorders—conditions affecting involuntary nervous system responses—and established the field's first specialized scientific journal.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) isn't

B) hasn't been

C) wasn't

D) weren't

17 Question 17

GRAMMAR

As one of aviation's earliest female stunt performers, Gladys Ingle developed a versatile repertoire of acrobatic feats. Her specialized skills included standing on the wing of a flying aircraft and shooting a bow and arrow at a _____ perfect balance during aerial maneuvers, such as vertical loops; and even walking across the wing as an aircraft flew at full speed beneath a bridge.

Which choice completes the text so that it conforms to the conventions of Standard English?

A target; maintaining

B target maintaining

C target. Maintaining

D target, maintaining

18 Question 18

GRAMMAR

During a recent investigation, archaeologists _____ an explanation for the consistent and seemingly deliberate placement of petroglyphs (rock carvings) near water analyzed several sites on Caribbean islands. They discovered that many of the pools near petroglyphs had consistent water levels year-round, creating perfect reflecting surfaces for the carved symbols on the rocks.

Which choice completes the text so that it conforms to the conventions of Standard English?

A had sought

B sought

C seeking

D were seeking

19 Question 19

GRAMMAR

For his mash-up “Smells Like Rockin’ Robin,” DJ Mark Vidler layered Kurt Cobain’s minor-key vocals from “Smells Like Teen Spirit” over the Jackson 5’s major-key blues progression from “Rockin’ _____ by inserting the grunge singer’s vocal melody as pitch-shifted “blue notes” within the pop-soul group’s harmonic framework, creates an intensified “bluesy” sound and subverts the expectations of listeners familiar with either musical source.

Which choice completes the text so that it conforms to the conventions of Standard English?

A Robin,” a technique that

B Robin.” A technique that

C Robin”; a technique that,

D Robin”—a technique that,

20 Question 20

GRAMMAR

During the late nineteenth century, agricultural chemist George Washington Carver revolutionized farming practices in the American _____ promoting systematic crop rotation techniques that replenished essential nitrogen in depleted soils.

Which choice completes the text so that it conforms to the conventions of Standard English?

A South, by

B South by

C South; by

D South: by

21 Question 21

GRAMMAR

Named for the year in which it was detected by English astronomers, SN 2016aps was a supernova (the explosion of a massive _____ in the constellation Draco, 3.6 billion light-years from Earth, the transient yet powerful blast propelled particles and debris into space at extremely high speeds.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) star). Occurring

B) star) occurring

C) star) and occurring

D) star), occurring

22 Question 22

GRAMMAR

Originating during the Cretaceous period, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

A) entomologists have extensively examined fossilized resin deposits excavated in northern Myanmar.

B) the fossilized amber specimens excavated in northern Myanmar preserve unprecedented anatomical details of early arthropods.

C) paleontological excavation in northern Myanmar has uncovered numerous ancient arthropods.

D) arthropods were trapped and entombed within sticky conifer secretions.

23 Question 23

GRAMMAR

In a 1993 study by Vales and Peek, the researchers used microhistological fecal analysis to determine the ratio of three plant subtypes (graminoids, forbs, and browse) within the diets of North American ungulates. The researchers did not perform this analysis on all such ungulates, _____ they focused exclusively on elk in Wyoming.

Which choice completes the text with the most logical transition?

A) for instance;

B) in other words;

C) regardless;

D) however;

24 Question 24

GRAMMAR

The Hornbostel-Sachs system classifies musical instruments by how they produce sound. For example, an instrument that is popular in India called the saraswati veena produces sound primarily through the vibration of its strings. _____ under the Hornbostel-Sachs system, the saraswati veena is a chordophone.

Which choice completes the text with the most logical transition?

A) Furthermore,

B) Thus,

C) Similarly,

D) Instead,

25 Question 25

GRAMMAR

Generally, sleek vehicles are more aerodynamic than bulkier ones. The streamlined nose of the Airbus A350 XWB jet, _____ helps it glide through wind with relative ease, while a boxy pickup truck encounters more wind resistance, making it less aerodynamic.

Which choice completes the text with the most logical transition?

A additionally,

B however,

C for example,

D meanwhile,

26 Question 26

GRAMMAR

India’s prominence as a supplier of information technology services to companies in North America and Europe is a result of several factors, including the country’s investments in technical education, but the most serendipitous is India’s large number of English speakers. English is spoken by more people in India—129 million—than in any nation except the US. _____ only about 10% of Indians speak English.

Which choice completes the text with the most logical transition?

A However, English is not the first language for most English speakers in India:

B Granted, this numerical advantage is due more to population size than to language penetration:

C As a result, careers in information technology services are not available to all people in India:

D Accordingly, 90% of people in India—over one billion people—do not speak the language:

27 Question 27

GRAMMAR

On June 13, 1927, aviator Charles Lindbergh received a ticker-tape parade in New York City in recognition of his transatlantic flight. Of the 206 ticker-tape parades held between 1886 and 2022, a notable number were for achievements in aviation. _____ Lindbergh's parade was just one of 16 honoring pilots.

Which choice completes the text with the most logical transition?

A Therefore,

B Indeed,

C In addition,

D Nevertheless,

28 Question 28

GRAMMAR

Between February 1876 and September 1876, George Eliot's novel *Daniel Deronda* was serialized in eight parts in *Blackwood's Magazine*. Scholars typically regard the serial version of Eliot's novel as less authoritative than the version later published in book form. _____ as scholar Philip Allingham notes, serialized texts were the versions in which the nineteenth-century readership first "saw itself and its world reflected, and to which it responded, often with great emotion."

Which choice completes the text with the most logical transition?

A Accordingly, they emphasize the appeal of the serial version for Victorian audiences;

B Still, the serial version is important to an understanding of the novel's historical context;

C Consequently, the book version of the novel is often favored over the serial version;

D However, others regard the serial version as a more faithful representation of Eliot's intentions;

29 Question 29

GRAMMAR

- The Highpointers Club is a hiking club.
- One of the main goals among club members is to reach the highest points in all fifty US states.
- Those who achieve this are called 50 Completers.
- In Suk Han became a 50 Completer on November 15, 2013.
- The highest point in Wisconsin is Timms Hill, at 1,951 ft.
- The highest point in Wyoming is Gannett Peak, at 13,809 ft.

Which choice most effectively uses information from the given sentences to explain the 50 Completers hiking challenge to a new audience?

A. If you are looking for a new hiking challenge, consider joining the Highpointers Club, as did In Suk Han, a hiker who successfully reached the highest point in every US state.

B. Not until after you have reached the highest points in all fifty US states—including Timms Hill in Wisconsin and Gannett Peak in Wyoming—can you include yourself among the 50 Completers of the Highpointers Club.

C. On November 15, 2013, In Suk Han finally completed the feat of reaching the highest point in all fifty US states, including Timms Hill in Wisconsin and Gannett Peak in Wyoming.

D. Hikers aiming to count themselves among the 50 Completers must reach not only Wisconsin's Timms Hill but also the even higher peak of Gannett Peak in Wyoming.

30 Question 30

GRAMMAR

While researching a topic, a student has taken the following notes:

- Enhanced rock weathering (ERW) is a method of spreading crushed basalt on farmland to capture carbon dioxide (CO_2) and improve soil health.
- ERW increases crop yields and removes harmful CO_2 from the air.
- ERW offers a low-cost, low-tech alternative to expensive, energy-intensive carbon capture methods.
- The CO_2 captured by ERW becomes stable bicarbonate in soil water, which eventually streams into the ocean to remain safely stored for hundreds of years.
- ERW can improve rural economies by reducing farmers' need for fertilizers and increasing their earnings from carbon credits.

The student wants to explain why ERW is a superior method for capturing CO_2. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A. ERW improves soil health and helps farmers increase their income from selling carbon credits.

B. ERW removes CO_2 from the air in a way that is both low-cost and low-tech.

C. The CO_2 captured by ERW becomes stable bicarbonate, which improves soil health by remaining in the ocean for hundreds of years.

D. ERW is a method of spreading crushed basalt on farmland to capture CO_2, improving rural economies.

Section 2, Module 1: Math

Baseline • 24 questions

1 Question 1

MATH

The function h is defined by $h(x) = 100x - 50$. What is the value of $h(x)$ when $x = 6$?

A 56

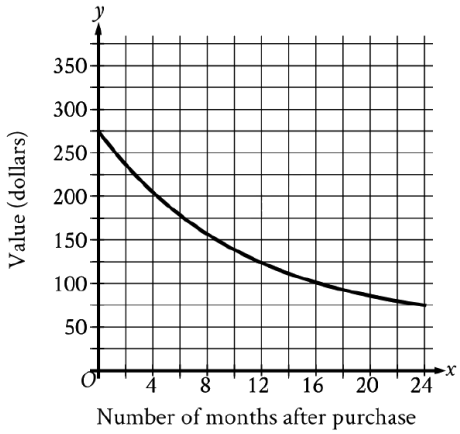
B 156

C 550

D 650

2 Question 2

MATH



The graph shown gives the estimated value, in dollars, of a mountain bike as a function of the number of months since it was purchased. What is the best interpretation of the *y*-intercept of the graph in this context?

A. The estimated value of the bike was $275 when it was purchased.

B. The estimated value of the bike 24 months after it was purchased was $275.

C. The estimated value of the bike had decreased by $275 in the 24 months after it was purchased.

D. The estimated value of the bike decreased by approximately 2.75% each year after it was purchased.

3 Question 3

MATH

Out of 200 attempts, 55% were successful. How many of these attempts were successful?

STUDENT-PRODUCED RESPONSE

Your answer: _____

4 Question 4

MATH

Which expression is equivalent to $29x^2 + 29$?

A) $29(x^2 + 1)$

B) $29(x^2 + 29)$

C) $29(29x^2 + 1)$

D) $29(x^2 + 1/29)$

5 Question 5

MATH

If $2(x - 3) = (2)(90)$, what is the value of $x - 3$?

A) 30

B) 60

C) 90

D) 180

6 Question 6

MATH

A commercial water pump drains water from a reservoir at a constant rate of 180 gallons per minute. At this rate, how many hours will it take to drain 32,400 gallons of water?

A) 2.5

B) 3.0

C) 3.5

D) 4.0

7 Question 7

MATH

A store sells two different-sized bottles of a certain shampoo. The store's sales of this shampoo totaled 902.07 dollars last month. The equation $4.57x + 6.08y = 902.07$ represents this situation, where x is the number of smaller bottles sold and y is the number of larger bottles sold. According to the equation, what is the price, in dollars, of each smaller bottle?

STUDENT-PRODUCED RESPONSE

Your answer: _____

8 Question 8

MATH

Triangle *ABC* is similar to triangle *XYZ*, where *A*, *B*, and *C* correspond to *X*, *Y*, and *Z*, respectively. In triangle *ABC*, the length of *AB* is 180 and the length of *BC* is 900. In triangle *XYZ*, the length of *YZ* is 80. What is the length of *XY*?

A) 288

B) 196

C) 80

D) 16

9 Question 9

MATH

$$|x| - 43 = 0$$

Which value is a solution to the given equation?

A) 1/43

B) 0

C) −1/43

D) −43

10 Question 10

MATH

In the *xy*-plane, line *k* passes through the points (0, −8) and (1, −5). Which equation defines line *k*?

A $y = (1/3)x - 8$

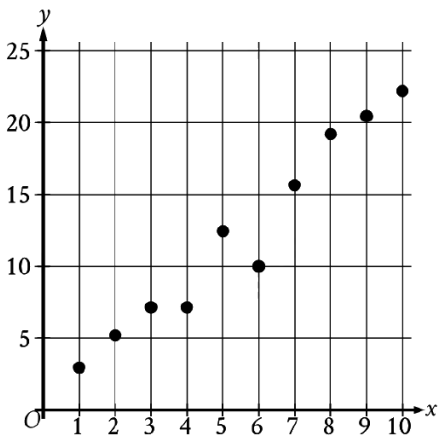
B $y = -x + 1/3$

C $y = 3x - 8$

D $y = -x + 3$

11 Question 11

MATH



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A $y = 0.2 + 2.25x$

B $y = 0.2 - 2.25x$

C $y = 2.25 + 0.2x$

D $y = 2.25 - 0.2x$

12 Question 12

MATH

What is the *x*-intercept of the graph of $y = (21 - x) / (ax + b)$ in the *xy*-plane, where *a* and *b* are positive constants?

A. $(21, 0)$

B. $(0, -21)$

C. $(0, 21/b)$

D. $(-b/a, 0)$

13 Question 13

MATH

A model estimates that a glacier moves between 17 and 24 centimeters per day. Which inequality represents possible values of *x*, where *x* is the total distance, in centimeters, the glacier moves in 2 days?

A. $0 \le x \le 2$

B. $(17)(0.5) \le x \le (24)(0.5)$

C. $17 \le x \le 24$

D. $(17)(2) \le x \le (24)(2)$

14 Question 14

MATH

In the ancient Roman measurement system, a leuga was a unit of length that was equal to 7,500 pedes. In this measurement system, how many pedes were equivalent to 4.6 leugas?

STUDENT-PRODUCED RESPONSE

Your answer: _____

15 Question 15

MATH

A rectangle has a width of n yards and a length of $(n + 7)$ yards. If the rectangle has an area of 198 square yards, what is the value of n?

STUDENT-PRODUCED RESPONSE

Your answer: _____

16 Question 16

MATH

$$55 = 2K / v^2$$

For an object with a mass of 55 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v = 2K / 55^2$

B) $v = 55^2 / 2K$

C) $v = \sqrt{(55 / 2K)}$

D) $v = \sqrt{(2K / 55)}$

17 Question 17

MATH

If $j/k = 83$ and $44j / (22xk) = 83$, what is the value of x?

A) 22/83

B) 1/2

C) 2

D) 44

18 Question 18

MATH

A certain company measures its solar cell production $C(n)$, in peak kilowatts, generated by all the solar cells manufactured by the company. The company's solar cell production grew by approximately 22% each year for the 5 years following 2013. If solar cell production was 7,138 peak kilowatts in 2013, which of the following functions best models the solar cell production, in peak kilowatts, n years after 2013, where $0 \le n \le 5$?

A. $C(n) = 7{,}138(0.22)^n$

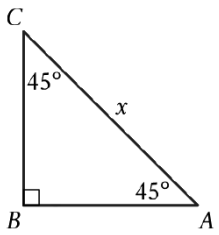
B. $C(n) = 7{,}138(1.22)^n$

C. $C(n) = 0.22(7{,}138)^n$

D. $C(n) = 1.22(7{,}138)^n$

19 Question 19

MATH



In the triangle shown, the value of x is 19. What is the value of $\sin A$?

A. $1/\sqrt{3}$

B. $1/\sqrt{2}$

C. 1

D. $\sqrt{2}$

20 Question 20

MATH

A right square prism has a height of 15 units. The volume of the prism is 1,815 cubic units. What is the length, in units, of an edge of the base?

STUDENT-PRODUCED RESPONSE

Your answer: _____

21 Question 21

MATH

$$y = x - 1$$

$$y = 19(x - 1)^2 - 20$$

A solution to the given system of equations is (x, y). What is one possible value of $x - 1$?

A. -2

B. $-20/19$

C. -1

D. $-1/19$

22 Question 22

MATH

$$18x + 6y = 8y + 9$$

$$ty = 1/5 + 9x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

STUDENT-PRODUCED RESPONSE

Your answer: _____

23 Question 23

MATH

A circle in the *xy*-plane has center (24, 5) and *y*-intercepts (0, −27) and (0, 37). What is the radius of this circle?

A) 40

B) 37

C) 32

D) 27

24 Question 24

MATH

For $x > 0$, which expression is equivalent to $\sqrt{(x^3)} \cdot \sqrt[4]{x}$?

A) $x^{3/8}$

B) $x^{7/4}$

C) $x^{5/4}$

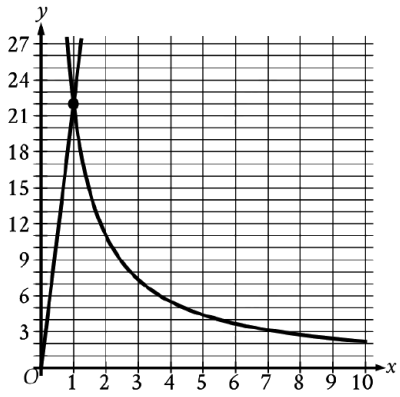
D) $x^{3/4}$

Section 2, Module 2: Math

Easier • 25 questions

1 Question 1

MATH



The system of two equations, $y = 22/x$ and $y = 22x$, is graphed in the xy-plane as shown. Which ordered pair is a solution to this system?

A (0, 0)

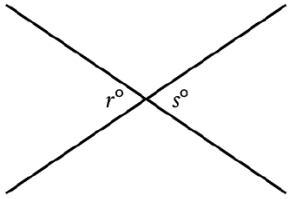
B (0, 22)

C (1, 22)

D (22, 22)

2 Question 2

MATH



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $r = 71$, what is the value of s?

A) 19

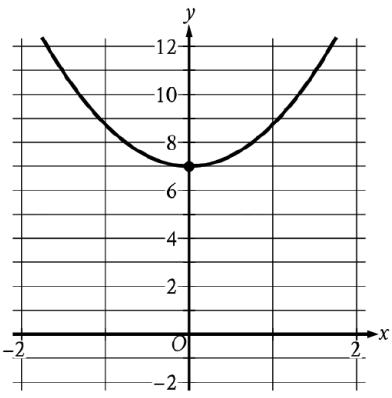
B) 71

C) 161

D) 251

3 Question 3

MATH



The y-intercept of the graph shown is (x, y). What is the value of y?

STUDENT-PRODUCED RESPONSE Your answer: _____

4 Question 4

MATH

$$x + y = 88$$

The given equation relates the total number of marble queen pothos stems, x, and the total number of snow queen pothos stems, y, a plant nursery propagates to create pothos plants. If 28 snow queen pothos stems are propagated, how many marble queen pothos stems are propagated?

A) 88

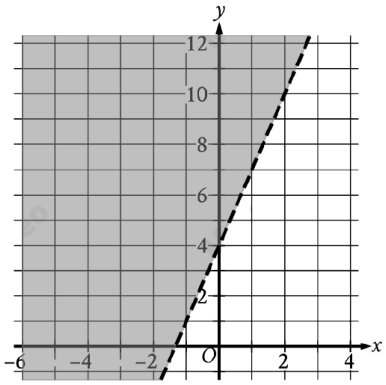
B) 60

C) 32

D) 28

5 Question 5

MATH



The shaded region shown represents the solutions to which inequality?

A $y < 3x + 4$

B $y < 3x - 4$

C $y > 3x + 4$

D $y > 3x - 4$

6 Question 6

MATH

If $3(2x - 5) = 4x + 7$, what is the value of x?

A 8

B 11

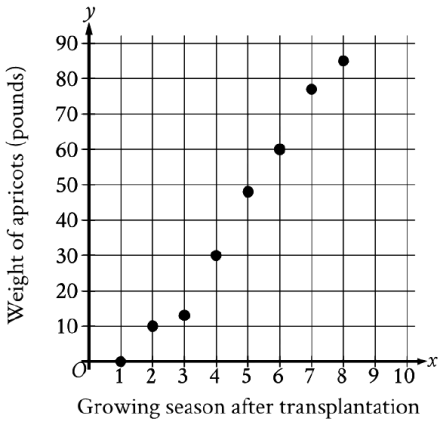
C 14

D 22

7 Question 7

MATH

A botanist recorded the weight, in pounds, of all apricots that grew on a dwarf apricot tree during each growing season after the tree's transplantation. The scatterplot shows this weight, in pounds, for each growing season after the tree's transplantation.



What was the weight, to the nearest pound, of all apricots that grew on the tree during the 4th growing season after the tree's transplantation?

STUDENT-PRODUCED RESPONSE Your answer: _____

8 Question 8

MATH

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 300 pounds. There are 13 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

A) 5

B) 10

C) 13

D) 34

9 Question 9

MATH

A data set consists of 8 integers: 4, 7, 7, 9, 12, 14, 15, 20. What is the median of this data set?

A) 7

B) 9

C) 10.5

D) 11

10 Question 10

MATH

Which expression is equivalent to $18x^2 - 3x^2$?

A) $-54x^2$

B) $15x^2$

C) $21x^2$

D) $42x^2$

11 Question 11

MATH

$$4(x - 5) + 8 = 2(x - 3) + 26$$

What value of x is the solution to the given equation?

A) 56

B) 32

C) 28

D) 16

12 Question 12

MATH

A market researcher collected data on the price of an iced coffee at 42 locations selected at random in a state. The mean price of an iced coffee in the state was estimated to be $1.37, with an associated margin of error of $0.05. Which of the following is a plausible statement about the mean price of an iced coffee for all locations that sell this product in the state?

A) It is between $1.32 and $1.42.

B) It is either less than $1.32 or greater than $1.42.

C) It is less than $1.32.

D) It is greater than $1.42.

13 Question 13

MATH

A teacher is creating an assignment worth 10 points. The assignment will consist of questions worth 1 point and questions worth 2 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 2-point questions?

A. $3xy = 10$

B. $3(x + y) = 10$

C. $2x + y = 10$

D. $x + 2y = 10$

14 Question 14

MATH

Data set X: 8, 12, 12, 16

Data set Y: 8, 12, 12, 16, 33

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

A. The mean of data set X is greater than the mean of data set Y.

B. The mean of data set X is less than the mean of data set Y.

C. The means of data set X and data set Y are equal.

D. There is not enough information to compare the means.

15 Question 15

MATH

The function f is defined by $f(x) = x^3 + 26$. What is the value of $f(2)$?

STUDENT-PRODUCED RESPONSE Your answer: _____

16 Question 16

MATH

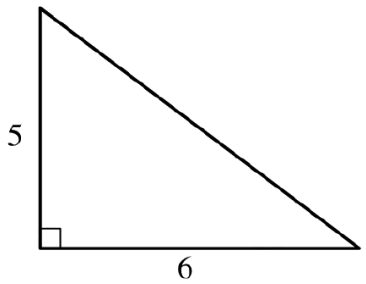
$$y = 2{,}200(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

STUDENT-PRODUCED RESPONSE Your answer: _____

17 Question 17

MATH



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

A) 15

B) 11

C) 6

D) 5

18 Question 18

MATH

What is the positive solution to the equation $x^2 - 7x - 18 = 0$?

A) 2

B) 6

C) 9

D) 18

19 Question 19

MATH

$$2x - 2y = 8$$

$$-3y = 18$$

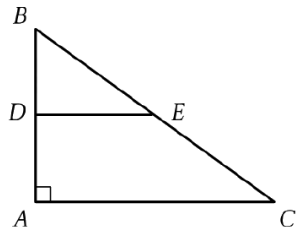
The solution to the given system of equations is (x, y). What is the value of x?

STUDENT-PRODUCED RESPONSE

Your answer: _____

20 Question 20

MATH



Note: Figure not drawn to scale.

In triangle *ABC*, *DE* is parallel to *AC*, *BD* = *AD*, *BE* = 9, and *AC* = 8. What is the length of *BC*?

A 72

B 18

C 17

D 16

21 Question 21

MATH

x	-4	0	8
y	11	12	14

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A $y = (1/4)x + 12$

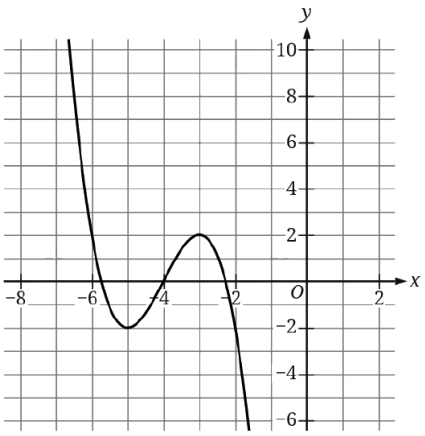
B $y = (1/12)x + 4$

C $y = 4x + 12$

D $y = 12x + 4$

22 Question 22

MATH



The graph of the function f is shown, where $y = f(x)$. What is the value of $f(-3)$?

A) -5

B) -3

C) 0

D) 2

23 Question 23

MATH

Line q is defined by $y = -(1/7)x - 1/2$. Line r is perpendicular to line q in the xy-plane. What is the slope of line r?

A) 9

B) 7

C) 5

D) 2

24 Question 24

MATH

$$y > 5x + 7$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A) x: 0, 1, 2 → y: 8, 13, 18

B) x: 0, 1, 2 → y: 7, 12, 17

C) x: 0, 1, 2 → y: 6, 11, 16

D) x: 0, 1, 2 → y: 5, 6, 7

25 Question 25

MATH

$$x^2 - 25/9 = 0$$

How many distinct real solutions does the given equation have?

A Zero

B Exactly one

C Exactly two

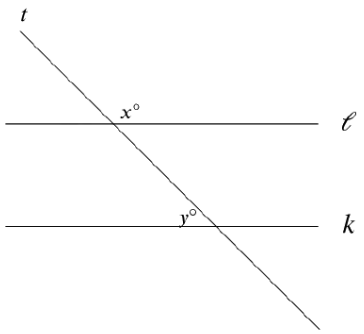
D Infinitely many

Section 2, Module 2: Math

Harder • 27 questions

1 Question 1

MATH



Note: Figure not drawn to scale.

In the figure shown, lines ℓ and *k* are parallel and line *t* intersects both lines. If $x > 137$, which of the following must be true?

A $y < 43$

B $y > 43$

C $x + y < 180$

D $x + y > 180$

2 Question 2

MATH

A circle in the *xy*-plane has equation $(x - 4)^2 + (y + 1)^2 = 49$. What is the radius of the circle?

A. 7

B. 14

C. 24.5

D. 49

3 Question 3

MATH

Area (square feet)	Water (gallons)
2,400	4,320
3,600	6,480
4,800	8,640

The roofs on the buildings of a public garden are designed to allow water to drain into the garden. The table shows the relationship between the area x, in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the garden over a certain period of time. Which equation could define f?

A. $f(x) = 0.6x$

B. $f(x) = 1.8x$

C. $f(x) = 2{,}160.0x$

D. $f(x) = 4{,}320.0x$

4 Question 4

MATH

$$15x + 435 = a(x + 29)$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

STUDENT-PRODUCED RESPONSE Your answer: _____

5 Question 5

MATH

What is the slope of the graph of $y = (1/3)(26x + 11) + 9x$ in the xy-plane?

A) 26/3

B) 53/3

C) 26

D) 35

6 Question 6

MATH

A plumbing company charges its customers \$116 for the first hour of a plumbing service call and \$70 for each additional hour, plus the cost of plumbing parts. Which equation represents the total cost y, in dollars, for an x-hour plumbing service call where the cost of parts was \$$z$ and $x \geq 1$?

A) $y = 70x + z$

B) $y = 70x + z + 46$

C) $y = 116x + z$

D) $y = 116x + 70z + 46$

7 Question 7

MATH

x	25	27	29
$f(x)$	−4	−4	4

For the quadratic function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f?

A $f(x) = (x - 25)(x - 27) + 2$

B $f(x) = (x - 25)(x - 27) - 4$

C $f(x) = (x - 4)(x - 29) + 16$

D $f(x) = (x - 4)(x - 29) - 16$

8 Question 8

MATH

The function h is defined by $h(x) = |x - 8| - |6 - x| + a$, where a is a constant, and $h(-4) = 1$. What is the value of a?

STUDENT-PRODUCED RESPONSE

Your answer: _____

9 Question 9

MATH

Line h is defined by $(1/3)x + (1/2)y - 30 = 0$. Line j is perpendicular to line h in the xy-plane. What is the slope of line j?

A. $-2/3$

B. $-3/2$

C. $2/3$

D. $3/2$

10 Question 10

MATH

A laboratory culture of bacteria grows according to the model $P(t) = 250(1.08)^t$, where t represents the elapsed time in hours. By what percentage does the population increase each hour?

A. 0.08%

B. 8%

C. 25%

D. 108%

11 Question 11

MATH

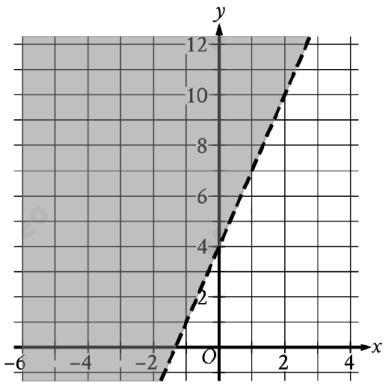
A beaker containing a liquid is placed on a table. The function $g(t) = 296 + (362 - 296)(2.72)^{-0.104t}$ gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

STUDENT-PRODUCED RESPONSE

Your answer: _____

12 Question 12

MATH



The shaded region shown represents the solutions to which inequality?

A) $y < 3x + 4$

B) $y < 3x - 4$

C) $y > 3x + 4$

D) $y > 3x - 4$

13 Question 13

MATH

$$3(2x) + 8(9y) = 534$$

$$3(2x) - 8(9y) = -618$$

The solution to the given system of equations is (x, y). What is the value of $2x + 9y$?

A) -86

B) -14

C) 58

D) 72

14 Question 14

MATH

The function f is defined by $f(x) = -2(x - 3)^2 + 8$. What is the maximum value of $f(x)$?

A) -2

B) 3

C) 6

D) 8

15 Question 15

MATH

The frequency table summarizes the age, in years, of each of the 50 coins in Vivek's original collection.

Age (years)	Frequency
60	10
65	5
70	4
75	12
80	4
85	5
90	10
Total	50

Vivek will add a 130-year-old coin to his original collection, creating a new collection of 51 coins. Which statement correctly compares the mean ages and median ages of Vivek's original and new collections?

A. The mean age for the new collection is greater than the mean age for the original collection, and the median age for the new collection is greater than the median age for the original collection.

B. The mean age for the new collection is greater than the mean age for the original collection, but the median age is the same for the original and new collections.

C. The mean age for the new collection is less than the mean age for the original collection, and the median age for the new collection is less than the median age for the original collection.

D. The median age for the new collection is greater than the median age for the original collection, but the mean age is the same for the original and new collections.

16 Question 16

MATH

$$x^2 + y^2 = 64$$

$$y = mx + b/2$$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point $(-7, y)$, where $y < 0$. Which expression represents the value of b?

A $14m - 2\sqrt{15}$

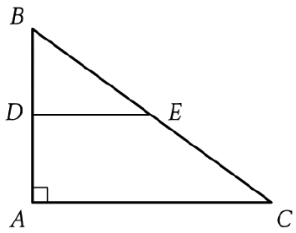
B $-14m + 2\sqrt{15}$

C $7m/2 - \sqrt{15}/2$

D $-7m/2 + \sqrt{15}/2$

17 Question 17

MATH



Note: Figure not drawn to scale.

In triangle *ABC*, *DE* is parallel to *AC*, *BD* = *AD*, *BE* = 9, and *AC* = 8. What is the length of *BC*?

A 72

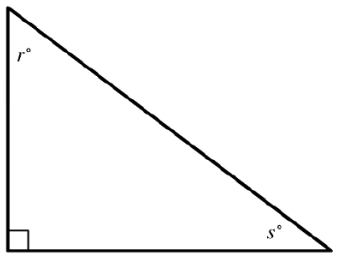
B 18

C 17

D 16

18 Question 18

MATH



Note: . igure not drawn to scale.

In the figure, $3r > 19s$. Which of the following values must be the greatest?

A) $\sin r°$

B) $\cos r°$

C) $\sin r° / \sin s°$

D) $\sin r° / \cos s°$

19 Question 19

MATH

$$2x + 3y = 19$$

$$x - y = 2$$

Consider the system of equations below. What is the value of $x + y$?

A 5

B 7

C 8

D 11

20 Question 20

MATH

A researcher conducted a study to examine the melatonin levels of gamers and the amount of time the gamers spent playing video games. A total of 125 gamers were selected at random from three countries. The melatonin levels of the gamers and the amount of time they spent playing video games were recorded over a 4-week period. The table gives the number of selected gamers from each country.

Country	Sweden	South Korea	United States
Number of gamers	25	49	51

If the results of this study suggest a relationship between melatonin levels and the amount of time spent playing video games, which of the following aspects of the study will prevent the results from being generalized to all gamers?

A. The gamers were not selected at random from all gamers.

B. The number of selected gamers is too small.

C. The number of selected gamers from each country is different.

D. The melatonin levels of the selected gamers were not measured over a long enough period of time.

21 Question 21

MATH

In right triangle ABC, angle C measures 90°. If $\cos(A) = 5/13$, what is the value of $\tan(A)$?

A. 12/5

B. 5/12

C. 12/13

D. 13/5

22 Question 22

MATH

At a hotel, guests used the pool for a total of *t* hours in March. At the same hotel, guests used the pool for 3.41*t* hours in April. What is the percent increase in the number of hours guests used the pool from March to April?

A) 2.41%

B) 3.41%

C) 241%

D) 341%

23 Question 23

MATH

$$-14(3x - 4)^2 + 6(3x - 2)^2$$

The given expression can be rewritten as $(a/4)x^2 + (b/4)x + c/4$, where a, b, and c are constants. What is the value of $a + b + c$?

A) −32

B) −16

C) −8

D) −2

24 Question 24

MATH

The sum of three consecutive even integers is less than 123. The least of the three integers is x. What is the greatest possible value of x?

A) 41

B) 40

C) 39

D) 38

25 Question 25

MATH

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 52 square centimeters (cm^2), and the surface area of right rectangular prism Y is 832 cm^2. The volume of right rectangular prism X is 10.5 cubic centimeters (cm^3). What is the volume, in cm^3, of right rectangular prism Y?

STUDENT-PRODUCED RESPONSE

Your answer: _____

26 Question 26

MATH

$$18(w - a)(w + 5) = 6(w + 5)$$

In the given equation, a is a constant. The sum of the solutions to the equation is 37/3. What is the value of a?

STUDENT-PRODUCED RESPONSE

Your answer: _____

27 Question 27

MATH

$$h(x) = a(x + 39)(x + r)(x + s)$$

The polynomial function *h* is defined by the given equation, where *a*, *r*, and *s* are positive constants. In the *xy*-plane, the graph of $y = h(x)$ intersects the *x*-axis at exactly three points. If $h(0) = p$, which of the following must be true?

A $p < 39rs$

B $p = 39rs$

C $p > 39rs$

D There is not enough information to compare the values of p and $39rs$.

Answer Key

Section 1, Module 1: Reading and Writing (Baseline)

1	2	3	4	5	6	7	8	9	10
B	D	B	C	A	B	D	D	A	C
11	12	13	14	15	16	17	18	19	20
D	A	C	A	C	A	A	B	B	C
21	22	23	24	25	26	27	28	29	30
D	B	B	B	C	D	D	B	B	B
31	32								
C	B								

Section 1, Module 2: Reading and Writing (Easier)

1	2	3	4	5	6	7	8	9	10
D	C	C	C	B	D	B	A	C	D
11	12	13	14	15	16	17	18	19	20
A	D	C	B	A	B	D	D	C	C
21	22	23	24	25	26	27	28	29	
C	A	C	A	D	B	A	B	C	

Section 1, Module 2: Reading and Writing (Harder)

1	2	3	4	5	6	7	8	9	10
D	A	C	B	A	D	D	B	D	D
11	12	13	14	15	16	17	18	19	20
A	D	B	C	B	D	A	C	D	B
21	22	23	24	25	26	27	28	29	30
A	B	B	B	C	B	B	B	B	B

Section 2, Module 1: Math (Baseline)

1	2	3	4	5	6	7	8	9	10
C	A	110	A	C	B	4.57	D	D	C
11	12	13	14	15	16	17	18	19	20
A	A	D	34,500	11	D	C	B	B	11
21	22	23	24						
C	1	A	B						

Section 2, Module 2: Math (Easier)

1	2	3	4	5	6	7	8	9	10
C	B	7	B	C	B	30	D	C	B
11	12	13	14	15	16	17	18	19	20
D	A	D	B	34	2,200	A	C	−2	B
21	22	23	24	25					
A	D	B	A	C					

Section 2, Module 2: Math (Harder)

1	2	3	4	5	6	7	8	9	10
A	A	B	15	B	B	B	−1	D	B
11	12	13	14	15	16	17	18	19	20
362	C	C	D	B	A	B	C	C	A
21	22	23	24	25	26	27			
A	C	A	D	672	17	D			